

SAMANVAY

An AI-enabled platform that integrates, harmonises, validates and synchronises multi-source land data — and shows its working.

PS 26013 · Smart India Hackathon 2026

Ministry of Rural Development · Department of Land Resources · Smart Automation

531,020 real features · 5 government sources · 76.7 km² of Chennai · zero synthetic data

THE PROBLEM

One wall. Six authoritative positions. All defensible.

PRODUCING AUTHORITY	WHAT IT PRODUCES	TYPICAL ERROR	CADENCE
Drone survey / photogrammetry	ORI, DSM, DTM	3–10 cm	one campaign
GeoAI feature extraction	building & parcel polygons	0.3–2 m	per campaign
Municipal corporation GIS	ward, road, property	1–5 m	continuous
Revenue department	cadastral FMB / survey numbers	5–30 m	decadal
Utility agencies	water, sewer, power	1–10 m	continuous
GNSS / CORS control	check points	1–3 cm	campaign

This is not an ETL problem. ETL assumes the right answer is in the sources and the job is to move it.
Here the sources disagree legitimately — and the job is to adjudicate, defensibly, at scale.

What SAMANVAY produces from that mess

1

One harmonised fabric

Parcels and buildings reconciled across every source, each with a ULPIN (Bhu-Aadhaar).

2

Explainable confidence

Six dimensions and an A-E grade: publish, desk-review, or send a field team.

3

An adjudication queue

Only genuinely contested cases reach a human — ranked by value, batched by cause.

4

Typed discrepancies

A subdivision, a re-survey and a department's missing data imply different actions.

5

A verifiable ledger

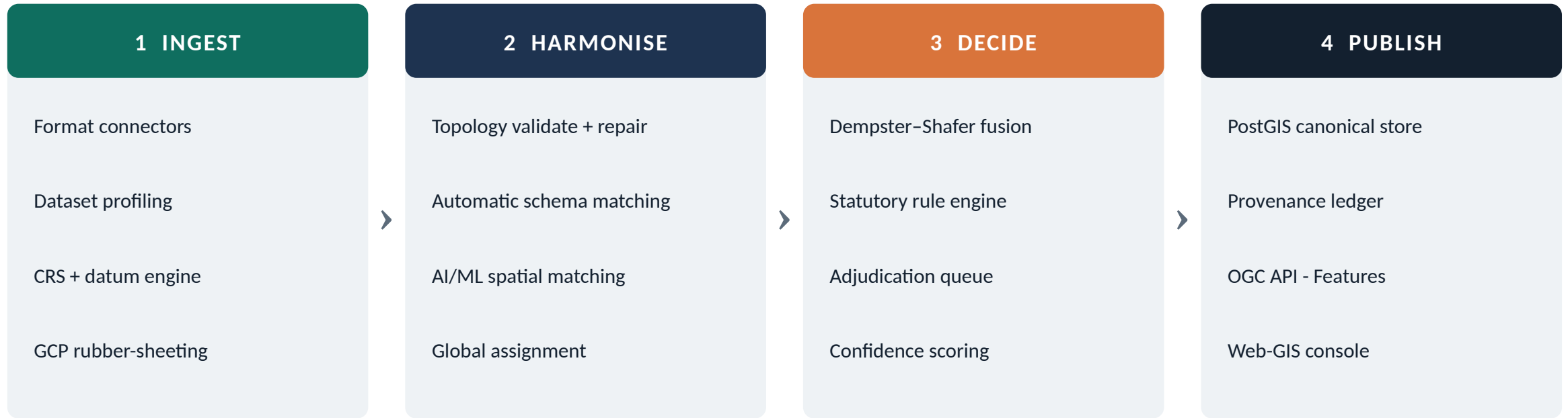
Every claim and decision hash-chained; a citizen can verify their own record.

6

OGC API - Features

Another department consumes it from QGIS with no SAMANVAY-specific code.

Four layers, one commitment: nothing is silently overwritten



Claims, not records

A claim is never deleted — it is superseded by a resolution that names it, states its belief and plausibility, and gives the reasoning in the language of a land record.

EVIDENCE

Real government data. No synthetic parcels anywhere.

DATASET	ISSUING AUTHORITY	EXTENT	LICENCE
TNGIS Tamil Nadu cadastrals	TNeGA (State)	6,017,242 parcels	CC0
NCSCM coastal cadastrals	MoEFCC	220,668 parcels	CC0
GCC Chennai building survey	Greater Chennai Corporation	964,053 footprints	CC0
Google Open Buildings v3	Google Research	15,889,026 buildings	CC-BY
AMRUT / Bhuvan buildings	NRSC · ISRO · MoHUA	2,163,113 footprints	CC0
UAV survey corpus (ORI + DSM)	OpenDroneMap	3 flights × 8 engines	CC-BY

10.7 M

features scanned across the national and state corpus

531,020

real features clipped to the Chennai AOI

76.7 km²

of central Chennai harmonised end to end

0

synthetic features, anywhere

Learning to match, with no labelled truth set in existence

1

Block

R-tree search radius derived from the two layers' declared accuracies — not a magic constant

2

Featurise

22 features: overlap, position, turning function, Hu moments, semantics, neighbourhood context

3

Register

Remove the systematic offset before matching, or IoU collapses for every pair at once

4

Weak-label

8 labelling functions; disagreement abstains rather than guesses

5

Self-train

The model labels what it is confident about; refit over the fuller distribution

6

Assign

Exact linear-sum assignment per component, then detect 1:N and N:1 groups

99.92%

of the 39 M possible pairs eliminated by blocking

1.000

cross-validated AUC on held-out weak labels

0.0003

expected calibration error

1,854

subdivisions and amalgamations detected

MEASURED ON REAL DATA

1.51 m

systematic offset between the Greater Chennai Corporation survey and the Google Open Buildings extraction, on a bearing of 073°

- Estimated from 5,226 confident pairs by the median, not the mean — the sample is contaminated by genuine change, and the median is unmoved by up to half of it.
- Removed before matching. Left in, it collapses IoU for every pair simultaneously and the matcher's most informative feature becomes noise.
- Reported to the department, because it is a finding about their data — not something to quietly correct and forget.

Why it matters

A 1 m offset across a ward is roughly the width of a set-back.

Overlay the two layers without removing it and every building appears to encroach slightly on its neighbour — thousands of false findings, each of which costs an officer's time to dismiss.

Remove it first and the residual disagreement is the real disagreement.

5,226 pairs • median estimator • under one second

Deciding between sources — and knowing when not to

What everyone tries, and why it fails

Trust the newest

A recent bad survey beats an old good one

Trust the highest authority

Revenue is authoritative on ownership, hopeless on geometry

Average the geometries

Invents a boundary no surveyor observed and none can re-verify

Majority vote

Three derivatives of one survey count as three witnesses

What SAMANVAY does

1

Statutory rules

A structure on poramboke land is an encroachment finding — not evidence the classification is wrong.

2

Domain precedence

Tenure from the revenue record. Position from GNSS/CORS. Ward from the corporation.

3

Evidence fusion

Dempster-Shafer with independence discounting, so shared-lineage sources count once.

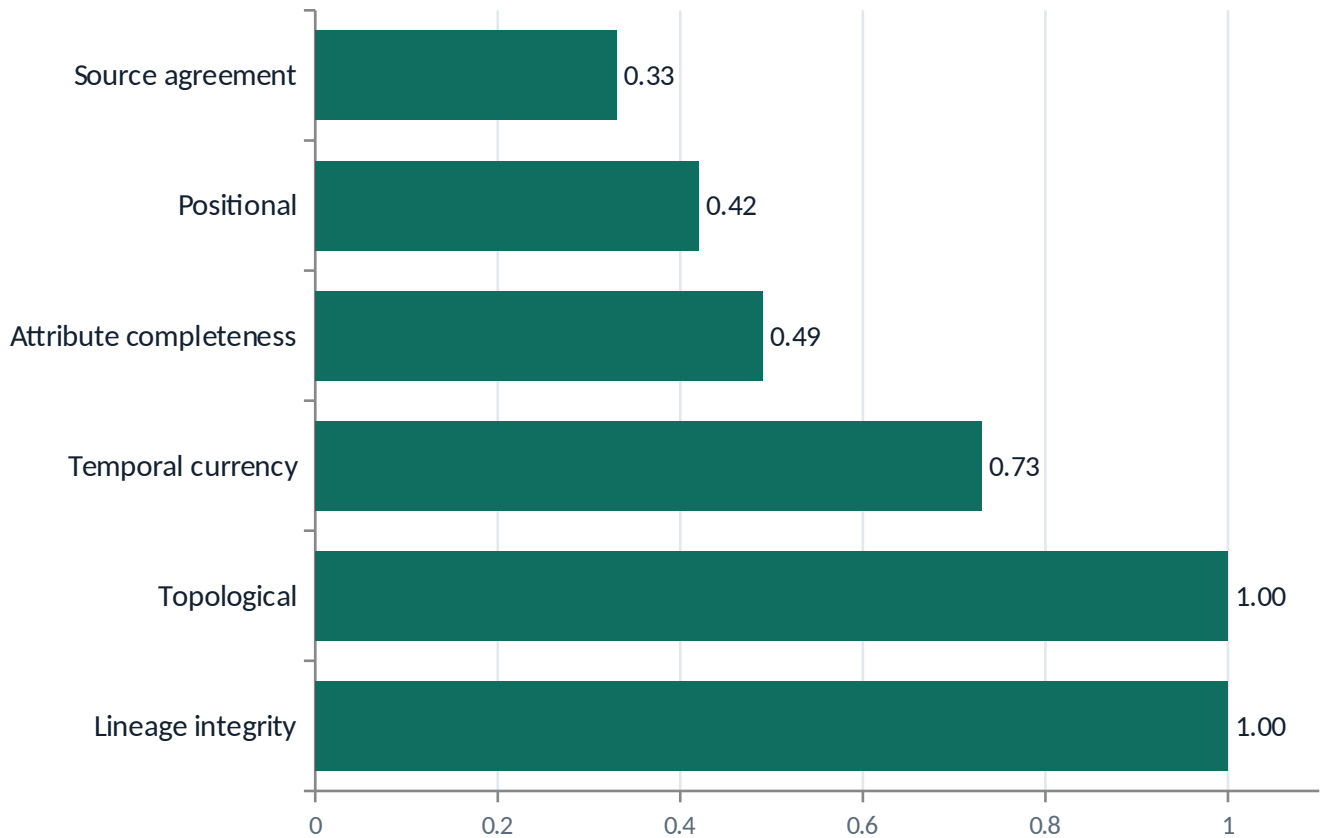
4

Escalation

High conflict mass, low belief or thin evidence → a human, with everything needed to decide.

The platform never averages geometries. Every boundary it emits is one that some survey actually observed.

Confidence an officer can act on



A-E grades, not a number

- A** Publish without review
- B** Publish, flagged
- C** Desk review
- D** Field verification
- E** Reject

Every feature also carries its six component scores and a plain-language sentence naming its weakest dimension.

Measured over 107,262 harmonised features across 22 km² of central Chennai. Positional is scored against the NAKSHA 0.5 m urban specification.

Two ways this saves an officer's day

The adjudication queue

43.4%

of conflicts resolved automatically

2

batches hold all 20,207 escalated cases

Cases are ranked by the expected value of deciding them — severity, uncertainty, land at stake, and how many similar cases the decision would settle — then grouped by cause, so an officer decides one kind of thing at a time.

Every decision becomes a training example and updates that source's empirical reliability, so the priors self-correct over a campaign.

The re-survey trap

A re-survey that moves every boundary in a village by 1.2 m is not two thousand mutations.

The detector estimates the layer-wide systematic offset, subtracts it, and classifies on the residual. Movement explained by the systematic component is typed POSITIONAL_ONLY and marked non-actionable.

Equally: contemporaneous sources are not epochs. Cross-source difference is a completeness finding for a custodian, never a mutation for a registry.

Each change type carries the registry action it implies — assess for tax, close the assessment, record a mutation, serve notice — in the language a revenue officer uses.

Drone imagery, ORI and DSM — measured, not asserted

ORI REBUILT

0.051 m

ground resolution, mosaicked from a real UAV tile pyramid into a georeferenced COG

CO-REGISTRATION

0.14 m

residual shift between two engines' reconstructions; peak sharpness 24

NULL EXPERIMENT

1.79%

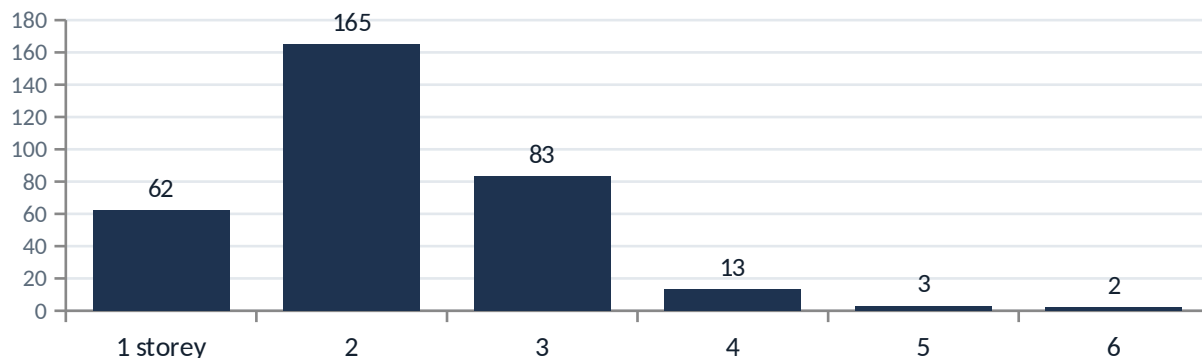
measured false-positive rate of the change detector on identical ground

DSM → DTM

0.02 m

median nDSM over ground — the terrain correctly sits at zero

Structures from height alone — no imagery, no labels, no model



328 structures segmented from the normalised surface model of a real 1 m airborne DSM. Median footprint 198 m², median height 6.43 m, storeys peaking at two — which is what the site actually looks like.

Regularisation then squares the wobbly raster outlines: 52.3% fewer vertices, mean area change -1.15%, and it refuses to square the 106 buildings that are genuinely not rectilinear.

Eight things the problem statement did not ask for



ULPIN minting

Bhu-Aadhaar identity stable under re-survey, unique by construction, checksum-protected



Verifiable ledger

Hash-chained with Merkle proofs — a citizen can verify their own record independently



Encroachment intelligence

Change detection on poramboke land, reported as findings for verification



Area reconciliation

Recorded vs geodesic extent, in acre-cent, ground, guntha, kanal



Active learning

Every adjudication updates that source's empirical reliability, Bayesian-style



LOD1 3-D city model

CityJSON from harmonised footprints and measured heights; FSI and coverage checks



Subscription bus

A utility agency is told the moment a parcel it depends on changes



DPDP-aware PII

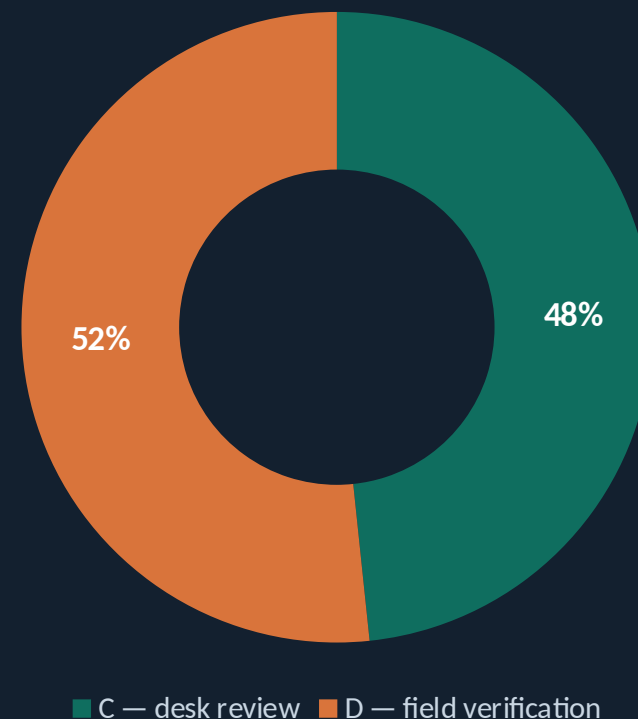
Owner data absent from the feature API; purpose-bound and ledger-logged

THE FINDING

Chennai's existing land data does not reach the NAKSHA specification — and now we can say by how much.

- No feature grades A or B against a 0.5 m positional specification.
- The best available source is a 1 m municipal survey.
- 68.0% of features are corroborated by only one source.
- Weakest dimension across the whole AOI: source agreement, at 0.33.

A harmonisation platform whose first run reported that everything was fine would not be measuring anything.



107,262 harmonised features by confidence grade

This is the map of where to fly the drones first.

Built to fit the frame it has to operate in

Conforms to

DILRMP

ULPIN, mutation genealogy, extent reconciliation

NAKSHA

Positional confidence scored against the urban specification

LGD

The canonical schema is built on the national code hierarchy

OGC API - Features

Conformant — usable from QGIS with no custom code

DPDP Act 2023

Owner data absent from the feature API by construction

CityGML LOD1 / CityJSON

3-D massing for FSI and coverage compliance

What it will not do

- Confer title. It harmonises spatial records; title is a legal determination.
- Determine encroachment. It puts the geometric evidence in front of the officer who does.
- Average geometries. Every emitted boundary was observed by some survey.
- Overwrite a source. Claims are immutable; resolutions supersede.
- Publish personal data through the feature API. Absent, not redacted on request.
- Claim measured accuracy it does not have. Without control it says so, per feature.
- Treat contemporaneous sources as epochs. Disagreement is not a mutation.

It runs.

Not a mock-up — a working platform, measured on real data.

~10,000

lines of Python across 30 modules

81

tests, all passing

107,262

features harmonised in one 17.8-minute run

2 vCPU

and 7 GB of RAM — district-office hardware

Every parcel it publishes carries its confidence, its contributing sources and the hash of the ledger entry that produced it. A record whose history cannot be reconstructed is a rumour with a geometry column.